Progressive Synergistic Registration and Fusion Diffusion Network for Unregistered Hyperspectral and Multispectral Image Fusion

Jiahui Qu, *Member, IEEE*, Jie He, Yunsong Li, *Member, IEEE*, Wenqian Dong, *Member, IEEE*, and Sen Liu

***Abstract*— Fusion of hyperspectral and multispectral images (HSIs and MSIs) has become a mainstream technique for obtaining high-spatial-resolution HSIs. However, most existing fusion methods are based on the ideal assumption that the low-resolution HSI (LR-HSI) and high-resolution MSI (HR-MSI) are perfectly registered, which is difficult to achieve in the practical scenario. Some works attempt to fuse unregistered images, treating registration and fusion as independent tasks; however, this will lead to the accumulation of registration errors in the fusion result. In this work, we propose a progressive synergistic registration and fusion diffusion network, called PSRF-DiffNet, to iteratively co-optimize image registration and fusion throughout the reverse process in a progressive bridging diffusion model. Specifically, a novel fine registration and fusion bridging (FRFB) model is designed, which models the image registration and fusion problem in a single model to greatly reduce the sensitivity of fusion performance to registration accuracy. In FRFB, the attention-enhanced-matching fine registration (AFR) model is proposed to generate registered images, on which, the spatial–spectral-coupled fusion (S^2CF) model performs fusion operation to generate high-quality hyperspectral images (HSIs). FRFB further allows direct connections from the state of the preceding FRFB to the current FRFB, leading to a contiguous refinement mechanism. Extensive experimental results indicate that our proposed PSRF-DiffNet could effectively deal with different types and degrees of nonrigid deformations and yield high-quality HR-HSIs, demonstrating better performance than other state-of-the-art baselines. The code is available at https://github.com/Jiahuiqu/PSRF-DiffNet**

***Index Terms*— Diffusion, hyperspectral image, non-rigid registration, super-resolution.**

I. Introduction

HYPERSPECTRAL image (HSI) is a data cube that contains hundreds of spectral bands [1]. By virtue of the rich spectral characteristics, multifarious significant applications [2], [3], [4] have been profited such as medical diagnosis, target detection, and image fusion. However, the tradeoff between spectral resolution and spatial resolution is inevitable in the hyperspectral imaging systems, high-spectral-resolution images are captured at the expense of spatial resolution, which leads to a significant reduction in detection accuracy for downstream tasks [5]. Therefore, in practical applications, it is desirable to obtain the latent HSI by fusing high-spatial-resolution images with high-spectral-resolution images. This allows the fused image to have a satisfactory resolution along both spatial and spectral dimensions.

At present, a large number of fusion algorithms have been proposed, which are mainly classified into traditional and deep learning-based methods.

The traditional methods can be categorized into four classes, including component substitution (CS), multiresolution analysis (MRA), Bayesian approaches, and matrix factorization [6]. CS-based methods separate spatial and spectral information from low-resolution HSIs (LR-HSIs) and replace spatial components with high-resolution MSIs (HR-MSIs). These methods include principal component analysis (PCA) [7], intensity–color saturation [8], and guided filtering. The MRA-based methods consist of smoothing filter-based intensity modulation (SFIM) [9], the MTF-generalized Laplacian pyramid (MTF-GLP) [10], and regression-based high-pass modulation [11], which extract high-frequency details from high-resolution HSIs (HR-HSIs) [12] and inject them into LR-HSIs to enhance the textural details. Despite the simplicity of implementation of CS- and MRA-based methods, they may lead to significant spectral distortion. The Bayesian approaches rely on Bayesian statistics to infer probabilistic parameters and design different Bayesian estimators for combining the co-registered paired MSIs and HSIs [13]. Coupled nonnegative matrix factorization (CNMF) [14] is a representative work of matrix factorization methods, which alternately unmixes the HS-MS images by NMF to estimate the spectral features of endmembers and the abundance maps with high resolution, respectively. Although some progress has been made with these methods, they are heavily dependent on artificial prior information, making it difficult to obtain optimal solutions to the fusion inverse problem.

In recent years, deep learning has promoted the advancement of hyperspectral and multispectral image fusion (HMIF) [15], [16], [17], [18], [19], [20], showing excellent feature extraction capabilities. Yang et al. [21] proposed PanNet, which introduces high-spectral-resolution images to preserve spectral information and trains the network parameters in the high-pass filtering domain to preserve spatial information.

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Wang et al. [22] proposed a novel conditional invertible network (DCINN) for image fusion, addressing limitations of existing methods like learning image mapping (LIM) and learning detail mapping (LDM). DCINN combines a decomposing module, an auxiliary network (ANet), and a conditional INN (CINN) to preserve details and achieve high performance without violating volume-preserving constraints. Yu et al. [23] proposed an unsupervised fusion spatial reconstruction network (UFSRnet) that addresses the limitations of HSI coverage by reconstructing missing regions and adaptively learning differences between heterogeneous data through a sensor radiation deviation correction (SRDC) module, demonstrating robust performance on both simulated and real datasets. Chen et al. [24] proposed a novel cyclic Transformer based on cross-modality spatial–spectral interaction, which integrates LR-HSIs with high-resolution multispectral images by exploiting diverse interaction modes to explore the similarity and complementarity among cross-modality features, achieving state-of-the-art performance both quantitatively and visually. Yu et al. [25] proposed HMPNet, an interpretable model-driven deep network for fusing hyperspectral, multispectral, and panchromatic images, which leverages a deep prior and unrolled optimization to achieve superior spatial and spectral quality, outperforming state-of-the-art methods in both visual and quantitative evaluations. Wu et al. [26] introduced a new implicit neural representation-based hyperspectral fusion function called the implicit neural feature fusion function, which overcomes the inherent shortcomings of vanilla INR, thereby solving the MSI and HSI fusion problem. Yang et al. [27] integrated tensor theory with deep learning and proposed an unsupervised deep tensor network for HMIF by jointly decoupling the LR-HSI and HR-MSI as the principal components of spectral and spatial modes and a sharing code tensor to describe the interaction among different modes. Liu et al. [28] proposed an end-to-end dual-branch spatial–spectral fusion network (DSFNet) designed to enhance HSI reconstruction. By jointly optimizing the multispectral filter array (MSFA) with the spectral sensitivity function (SSF) response model and the DSFNet framework, the proposed method achieves a significant improvement in the reconstruction accuracy of HSIs. Cao et al. [29] proposed a novel supervised diffusion model with two conditional modulation modules—coarse-grained style modulation (CSM) and fine-grained wavelet modulation (FWM)—designed to disentangle style and frequency information for high-quality fused images. Realizing that CNNs have difficulty in capturing long-term dependencies in images, the Transformer-based model aims at modeling long-distance dependencies through a self-attentive mechanism. Hu et al. [30] designed a Transformer-based architecture, which can globally explore the intrinsic relationship within features. Fang et al. [31] presented a novel approach, the deep unfolding network enhanced by Transformer priors, which integrates the deep unfolding technique and capitalizes on the principles of mutual information (MI) and the power of deep learning priors to enhance the degradation model-informed fusion process, thereby achieving superior image fusion results. Sun et al. [32] proposed the MDC-FusFormer network, which effectively fuses spatial–spectral features of HSIs and MSIs at multiple scales, enhancing the quality of the fused images. Deng et al. [33] proposed a novel pyramid shuffle-and-reshuffle Transformer, which incorporates shuffle-and-reshuffle modules to efficiently consider the information interaction among global patches for HMIF. The advantage of deep learning-based methods is that all parameters in the network can be updated under the supervision of training samples, thus reducing the need for artificially designed prior. Therefore, deep learning is gradually becoming the mainstream method for image fusion.

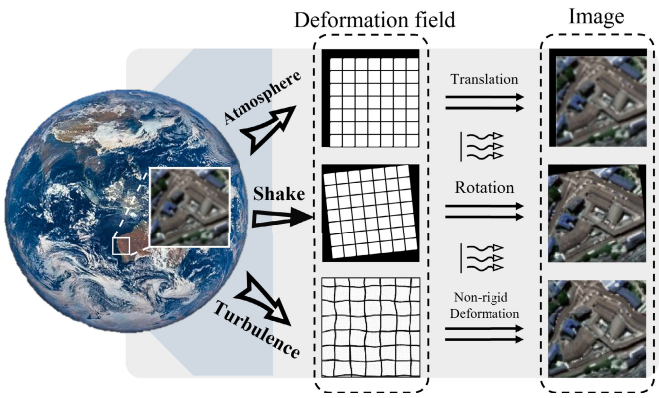


Fig. 1. Impact of terrain undulations and atmospheric turbulence on image formation, including translation, rotation, and nonrigid deformation.

Nevertheless, it is essential to note that HMIF highly relies on an ideal assumption of perfect registration between LR-HSIs and HR-MSIs, a condition that is seldom achievable in real-world scenes due to interference during the image acquisition process. Spatial distortion in HSI transmission results from multiple factors. In geospatial correction, before image acquisition, standard target imaging determines sensor geometric parameters for pixel positioning. After the acquisition, ground control points, satellite orbit parameters, and mathematical models correct geometric distortions, while digital elevation model data is used to consider terrain for accurate surface reflectance. In atmospheric correction, before acquisition, an appropriate atmospheric model is chosen based on location and season, and aerosol optical thickness is estimated. After the acquisition, the radiative transfer model is used to retrieve true surface reflectance and perform interband correction. Although geometric and atmospheric corrections are effective in rectifying large-scale rigid transformations, nonrigid transformations, which are induced by factors such as terrain irregularities, object-specific warping, and the complex interactions between the imaging system and the dynamic environment, pose distinct challenges that necessitate alternative solutions. Fig. 1 shows several common deformations during the imaging process of HSIs. While employing fusion methods designed for registered images to fuse unregistered ones, the performance will be subjected to a severe decline which can be attributed to the influence caused by mismatches and differences between the unregistered images. Therefore, it becomes imperative to exploit fusion algorithms tailored for unregistered hyperspectral and multispectral images (HSIs and MSIs).

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Recently many studies have been proposed to address the unregistered problem in HSIs and multispectral image fusion (UHMF) [34], [35], [36], [37], [38]. In these studies, according to the deformation types of input image pairs, the corresponding image registration strategy [39], [40] is added before image fusion to achieve accurate registration. For instance, Chen et al. [41] developed the SIRF method, which innovatively introduces dynamic gradient sparse features into the fusion process, effectively addressing the joint optimization of registration and fusion. Lin et al. [42] used spectral sparsity to recover unregistered high-resolution HSIs, while spectral and spatial structural correlation were used to recover high-resolution HSIs in the registered region. Fu et al. [43] proposed a method that allows for simultaneous HSI fusion and geometric registration of image pairs with highly contrasting spatial resolution, and they evaluated the method under five different geometric deformations. Xie et al. [44] proposed the SA-DNet framework, which adopts a two-stage strategy for registration and fusion: first performing feature matching by extracting region-of-interest features, followed by fusing the registered images using an image fusion module. Nie et al. [45] introduced spatial transform networks (STNs) to learn the parameters of affine transformations between input image pairs and then mapped the STNs and super-resolution networks into an unsupervised and weighted joint learning model for fusion and registration. However, these methods can only be used to process image pairs with affine transformation or rigid deformation. To deal with nonrigid deformations, Gao and Zheng [46] proposed ACSGRegNet, which generates precise deformation fields through a cross-attention mechanism, achieving end-to-end registration and fusion. Shi et al. [47] introduced a novel dual-parallel feature extraction architecture, significantly improving fusion performance through progressive feature space alignment and attention-guided cross-fusion. Hu et al. [48] proposed an unsupervised blind HSI super-resolution (UBHSI-SR) method. Designing dual unmixing branches with a shared decoder leverages an abundance of similarity constraints between LR-HSIs and transformed MSIs to guide the learning process. Li et al. [49] proposed an innovative framework (MulFS-CAP) for a single-stage fusion of unregistered infrared–visible images. This network constructs a correlation matrix to meticulously delineate pixel relationships between the source images to facilitate the alignment of features across infrared and visible spectra. Zhou et al. [50] proposed a registration method by minimizing the least-squares objective function using the point spread function (PSF). After registration, low-dimensional manifold invariant with local linear transformations is exploited for HSI fusion [51]. To capture the nonlinear statistical dependencies of two input data, Qu et al. [52] proposed the unsupervised MI Dirichlet network to solve the HSI super-resolution problem without registration. Zheng et al. [53] proposed a new unsupervised registration network, NonRegSRNet, which integrates fusion and dense resolution into a unified model. Guo et al. [54] simulated the structure of stereo cameras to propose a stereo cross-attention network to achieve an accurate fusion of unregistered LR-HSIs and HR-MSIs.

In summary, existing HMIF methods assume perfect registration between the input image pairs. However, imaging and transmission processes often introduce challenges due to the complex interaction between the imaging system and dynamic environments. Recently, researchers have begun addressing this issue by prioritizing registration before fusion. Nevertheless, treating registration and fusion as separate tasks results in error accumulation during the fusion process. Although some studies have attempted to jointly optimize fusion and registration tasks, these methods typically model the deformation between image pairs as either rigid or nonrigid transformations alone. Due to the complexity of the imaging process, such single-type deformation limits the improvement of registration performance to some extent. Against this backdrop, this article focuses on the joint optimization of registration and fusion, taking into account the diversity of image deformations. The registration process is modeled as a combination of coarse-to-fine rigid and nonrigid registration, thereby minimizing the adverse effects of combined rigid and nonrigid registration errors on the fusion process and ultimately enhancing the quality and accuracy of the fused images.

In this article, considering the interdependence of fusion and registration, we propose a progressive synergistic registration and fusion diffusion network (PSRF-DiffNet), which performs the registration and fusion task from coarse to fine in the form of diffusion. As shown in Fig. 2, we ingeniously combine the registration and fusion with the diffusion model, utilizing its iterative characteristics to implement a step-wise fusion and registration strategy. In each iteration, the fine registration and fusion bridging (FRFB) model is designed to achieve fine registration and generate the fusion image of the current stage, the result will be used as the input of the next iteration. The FRFB generates deformation fields for registration through a local matching strategy and then considers both spatial and spectral dimensions to maximize the preservation of the original spectral characteristics while significantly enhancing the spatial features. Multiple iterations were conducted until a high-quality fused image was obtained, with precise alignment of spatial information and significant improvement in spatial–spectral quality. The contributions of this article are as follows.

1) In this article, we propose an end-to-end PSRF-DiffNet. PSRF-DiffNet models registration and fusion as a complementary execution process, which achieves HSI fusion step-by-step through iterative registration and fusion of LR-HSIs and HR-MSIs.
2) We design an attention-enhanced-matching fine registration (AFR) model to perform registration. AFR generates local offset values by assessing the spatial–spectral similarity between HSIs and MSIs and subsequently constructs a deformation field to achieve fine registration of HSIs.
3) We propose a spatial–spectral-coupled fusion (S^2CF) model to fuse images. S^2CF enhances the texture details of the image by calculating the advanced version of the structural tensor (AST) and then injects the enhanced texture features into the spectral enhancement branch through guided filtering to achieve image fusion.

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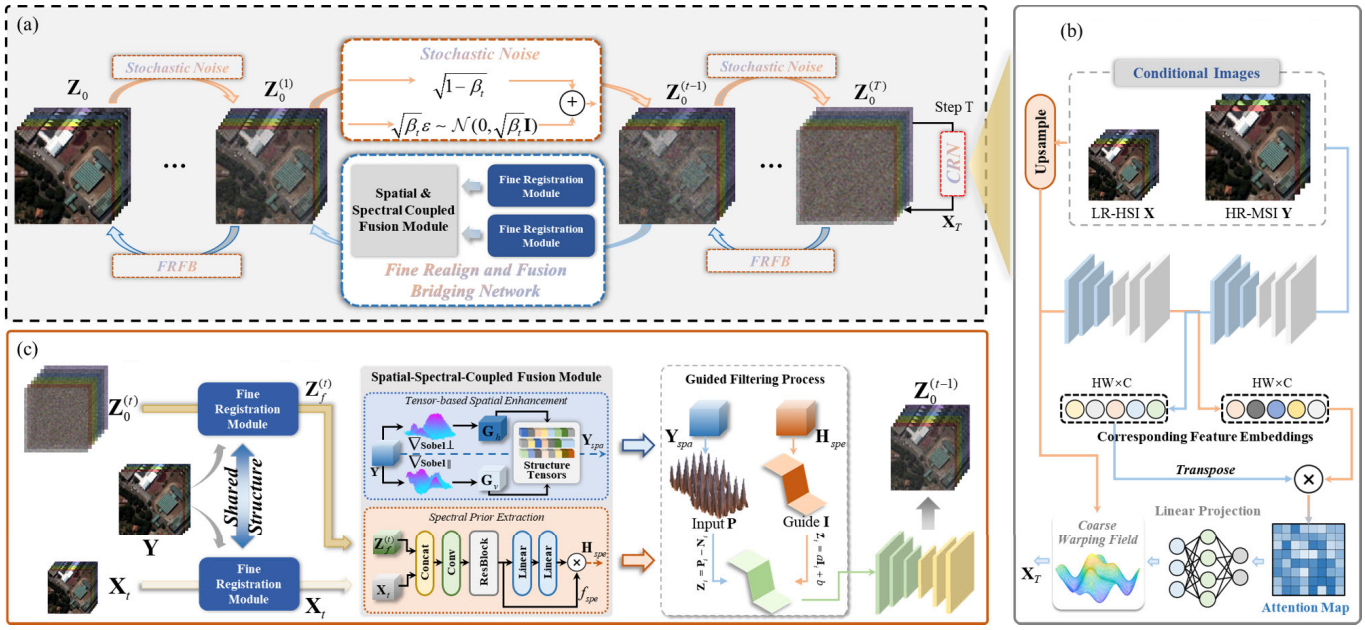


Fig. 2. Illustration of PSRF-DiffNet. (a) PSRF-DiffNet. (b) CRN. (c) Fine realign and fusion bridging network.

We arrange the remainder of this article as follows. The proposed PSRF-DiffNet is introduced in Section II. In Section III, we present qualitative and quantitative results for three datasets with different deformations and perform the ablation studies. The conclusion is drawn in Section IV.

II. Proposed Approach

In this section, we will introduce the proposed PSRF-DiffNet in detail. First, we model unregistered HMIF mathematically for intuitive knowledge. Then, in the diffusion model framework, we design CRN to roughly align source images, AFR for precise alignment, and S^{2}CF to combine the registered ones. By leveraging the progressive recovery features of the reverse reconstruction process of the diffusion model, we iteratively optimize outcomes to obtain a well-registered high-quality HR-HSI.

A. Problem Formulation

In this article, the problem we focus on is how to fuse HSIs and other data sources that have not been accurately registered to obtain high-resolution HSIs (HR-HSIs).

Given the fixed image HR-MSI $\mathbf{Y} \in \mathbb{R}^{c\times H\times W}$, the moving image LR-HSI $\mathbf{X} \in \mathbb{R}^{C\times h\times w}$, and the ideal image HR-HSI $\mathbf{Z} \in \mathbb{R}^{C\times H\times W}$, where $\{C \ll c\}$, $\{W \ll w\}$, and $\{H \ll h\}$ denote the number of bands, widths, and heights, respectively. Therefore, the spectral relationship between $\mathbf{Z}$ and $\mathbf{Y}$ can be expressed as

$$\mathbf{Y} = \mathbf{RZ} \tag{1}$$

where $\mathbf{R} \in \mathbb{R}^{c\times C}$ denotes the spectral response function.

Since atmospheric disturbance is inevitable during the imaging process, there are some geometric distortions contained in the acquired LR-HSI $\mathbf{X}$ so that a conflict occurs that it is hard to maintain a completely consistent spatial correspondence between $\mathbf{Z}$ and $\mathbf{X}$. The spatial relationship between $\mathbf{Z}$ and $\mathbf{X}$ can be represented as follows:

$$\mathbf{X} = (\mathbf{Z} \otimes \mathbf{G})\mathbf{B} \tag{2}$$

where $\mathbf{B} \in \mathbb{R}^{HW\times hw}$ is the point spread function consisting of the spatial blurring kernel and downsampling operation and $\mathbf{G}$ means the deformation function caused by the interference, such as terrain undulations and atmospheric turbulence.

The existing deep learning-based unregistered HSI fusion methods tend to regard this mission as a two-step "fusion after registration" process, which can be described as the following equation:

$$\mathbf{Z} = \mathcal{F}_\theta(\mathcal{G}(\mathbf{Y}, \mathbf{X}), \mathbf{Y}) \tag{3}$$

where $\mathcal{G}$ is the registration operation to reverse the deformation performed on $\mathbf{X}$ and $\mathcal{F}_\theta(\cdot)$ represents the fusion network parameterized by θ. Because this kind of approach operates registration and fusion separately, the computing errors generated during the registration process will be accumulated in the subsequent fusion step.

To this end, we introduce a novel architecture called the PSRF-DiffNet, which alternately performs registration and fusion on the LR-HSI and HR-MSI through a series of progressive iterations. The overall framework of the proposed PSRF-DiffNet is shown in Fig. 2. In the forward process of PSRF-DiffNet, the Gaussian noise is incrementally added to the HR-HSI until the image is entirely degraded into pure noise. In particular, a coarse registration network (CRN) is designed before the backward process to effectively capture the macroscopic correspondences between image pairs and to facilitate preliminary spatial transformations, enabling it to enhance the network's ability to concentrate on refining deformation fields and executing fusion tasks with greater precision. In the backward process, the HR-HSI can be reconstructed

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by alternately executing the fusion and registration, conditioned on LR-HSIs and HR-MSIs through multiple iterative time steps. In each time step, we propose the FRFB model, which performs fine registration and fusion successively by the AFR model and S^{2}CF model. The proposed formula for PSRF-DiffNet is as follows:

$$\mathbf{Z}_{t-1} = \phi(f_{\theta,t}[\mathcal{D}_t(\mathbf{Y}, \mathbf{X}_t), \mathbf{Y}, \mathcal{D}_t(\mathbf{Z}_t, \mathbf{Y})], \mathbf{Z}_t) \tag{4}$$

where $t \in [0, T]$ represents the tth time step, $\mathbf{X}_t$ is the registered LR-HSI, $\mathbf{Z}_t$ can be considered as the rough fusion result generated from the tth time step, $\mathcal{D}_t$ is fine registration methods, $f_{\theta,t}[\cdot]$ takes $\mathbf{X}_t$ and $\mathbf{Y}$ as guidance to refine $\mathbf{Z}_t$ at a higher resolution level by S^{2}CF, and $\phi(\cdot)$ is the image generation function that generates the fusion result of this iteration $\mathbf{Z}_{t-1}$.

Finally, the ideal reconstructed image can be obtained after T-step alternate iterations between the AFR and the S^{2}CF.

B. Overall Architecture of the PSRF-Diff

Given a clean HR-HSI $\mathbf{Z}_0$, the forward diffusion process gradually adds the Gaussian stochastic noise to $\mathbf{Z}_0$ through a Markov chain to generate the noisy image $\mathbf{Z}_T$

$$q(\mathbf{Z}_{1:T}|\mathbf{Z}_0) = \prod_{t=1}^{T} q(\mathbf{Z}_t|\mathbf{Z}_{t-1}) \tag{5}$$

$$q(\mathbf{Z}_t|\mathbf{Z}_{t-1}) = \mathcal{N}\left(\mathbf{Z}_t; \sqrt{\alpha_t}\mathbf{Z}_{t-1}, 1-\alpha_t\mathbf{I}\right) \tag{6}$$

where the time step $t \in [0, T]$ and the scalar parameters α_t $(0 < \alpha_t < 1)$ are hyperparameters, which determine the variance of the noise added at each iteration. Furthermore, we can directly derive $\mathbf{Z}_t$ from $\mathbf{Z}_0$ using the reparameterization technique

$$q(\mathbf{Z}_t|\mathbf{Z}_0) = \mathcal{N}\left(\mathbf{Z}_t; \sqrt{\gamma_t}\mathbf{Z}_0, (1-\gamma_t)\mathbf{I}\right) \tag{7}$$

$$\mathbf{Z}_t = \sqrt{\gamma_t}\mathbf{Z}_0 + \sqrt{1-\gamma_t}\varepsilon \tag{8}$$

where $\gamma_t = \prod_{i=1}^{T} \alpha_i$ and $\varepsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.

To improve the backward process's ability to focus on learning local fine deformations and to improve the robustness of image fusion, we first introduce CRN which performs a simple rigid coarse registration on the image pairs $\{\mathbf{X}, \mathbf{Y}\}$ before initiating the backward process. This step yields a roughly registered image $\mathbf{X}_T$, serving as the initial condition for the subsequent processing

$$\mathbf{X}_T = \mathrm{CRN}(\mathbf{X}, \mathbf{Y}) \tag{9}$$

where T is the final time step, and we treat the registered $\mathbf{X}_T$ as the conditioned LR-HSIs for the subsequent steps in the reconstruction process.

For the backward process, it can be formulated by Bayesian posterior distribution

$$p(\mathbf{Z}_{t-1}|\mathbf{Z}_t, \mathbf{Z}_0) = \mathcal{N}(\mathbf{Z}_{t-1}; \mu(\mathbf{Z}_t, \mathbf{Z}_0), \Sigma(\mathbf{Z}_t, \mathbf{Z}_0)) \tag{10}$$

where $\mu(\mathbf{Z}_t, \mathbf{Z}_0)$ is the mean and $\Sigma(\mathbf{Z}_t, \mathbf{Z}_0)$ denotes the variance, then we can obtain the ideal HR-HSI $\mathbf{Z}_0$ by inferring $\mathbf{Z}_0^{(t)}$ at the tth step with the FRFB model

$$\mathbf{Z}_0^{(t-1)} = \mathrm{FRFB}\left(\mathbf{Z}_0^{(t)}, \mathbf{X}_t, \mathbf{Y}\right) \tag{11}$$

where $\mathbf{Z}_0^{(t)}$ is the partially denoised HSI, $\mathbf{X}_t$ is the registered LR-HSI output from the tth $(t \in [0, T])$ step, and $\mathbf{Y}$ is the auxiliary HR-MSI.

Specifically, FRFB consists of two modules in each step: the AFR model and the S^{2}CF model. AFR is engineered to improve the precision of fine registration by leveraging advanced attention mechanisms, while S^{2}CF integrates spatial and spectral information to facilitate effective fusion. Collectively, these modules operate in a complementary manner to refine the registration and fusion processes, thereby ensuring high-quality outputs at each iterative step.

Despite the preliminary rigid registration of the LR-HSI with the HR-MSI, complex transformations resulting from nonrigid deformations still exist in the starting condition $\mathbf{X}_T$ which could impact the accuracy of the following operations. Therefore, AFR aims to perform a detailed registration on both the noisy image and LR-HSIs with HR-MSIs by generating a deformation field to register local mismatched areas. The precise registration process can be represented as

$$\mathbf{Z}_f^{(t-1)} = \mathrm{AFR}\left(\mathbf{Z}_0^{(t)}, \mathbf{Y}\right) \tag{12}$$

$$\mathbf{X}_{t-1} = \mathrm{AFR}(\mathbf{X}_t, \mathbf{Y}) \tag{13}$$

where $\mathbf{Z}_f^{(t-1)}$ and $\mathbf{X}_{t-1}$ are results of $\mathbf{Z}_0^{(t)}$ and $\mathbf{X}_t$ $(t \in [0, T])$ after fine registration through AFR. Specifically, the input LR-HSI at the first step is the roughly registered $\mathbf{X}_T$ from (9).

With the refined registration at each time step, we design S^{2}CF to enhance the spatial details and improve the spectral fidelity of the $\mathbf{Z}_f^{(t-1)}$ so that the high-quality $\mathbf{Z}_0^{(t-1)}$ can be obtained by fusing $\mathbf{Z}_f^{(t-1)}, \mathbf{X}_{t-1}, \mathbf{Y}$. The fusion process can be represented as

$$\mathbf{Z}_0^{(t-1)} = \mathrm{S}^2\mathrm{CF}(\mathbf{Z}_f^{(t-1)}, \mathbf{X}_{t-1}, \mathbf{Y}). \tag{14}$$

Subsequently, $\mathbf{Z}_0^{(t-1)}$ will be fed into the next iteration to undergo the same procedure as previously described. The final desired image $\mathbf{Z}_0$ will be generated after completing T time steps.

C. Coarse Registration Network

To reduce the difficulty of learning and inferring the fine deformation fields and improve the robustness of image fusion during the backward process, we design the CRN to generate a set of affine transformation parameters to perform macroscopic coarse alignment on LR-HSIs and HR-MSIs so that most of the pixels can be effectively matched. The specific procedure of coarse registration can be expressed as follows:

$$\begin{aligned}\mathbf{X}_T &= \varphi_\theta(\mathbf{X}, \mathbf{Y}) \\ &= \mathrm{Linear}\left[\mathrm{Softmax}\left(\frac{Q_{\mathbf{Y}}(K_{\mathbf{X}})^T}{\sqrt{C}}\right)\right]\end{aligned} \tag{15}$$

where $\mathbf{X}_T$ is the result of LR-HSI $\mathbf{X}$ after the rigid registration via the affine transformation function $\varphi_\theta(\cdot)$ and the beginning condition for the subsequent backward process, and the query $Q_{\mathbf{Y}}$ and the key $K_{\mathbf{X}}$ can be acquired through a linear projection layer.

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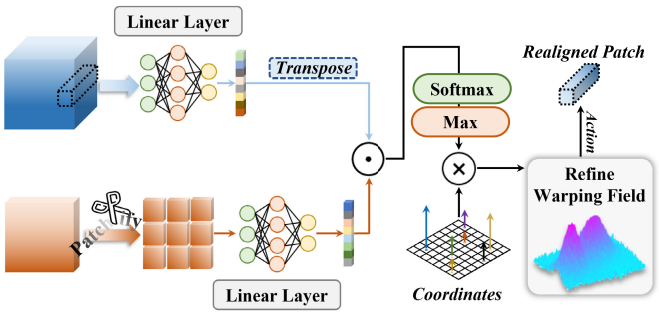


Fig. 3. Illustration of AFR model.

D. FRFB Network

1) AFR Model: Although LR-HSI $\mathbf{X}_T$ has been preliminarily registered by CRN which can only handle simple rigid deformation, we still have to confront those distortions caused by nonrigid deformation. Consequently, to obtain more precise registered images, the AFR model is proposed to generate a deformation field to register local unmatched patches and the upcoming fusion operation can be performed on a more reliable foundation within each step, thereby minimizing error accumulation and reducing overall discrepancies. Fig. 3 shows the framework of the proposed AFR. The brief description of this accurate registration can be expressed as follows:

$$\begin{aligned}\mathbf{X}_{t-1} &= \mathrm{AFR}(\mathbf{X}_t, \mathbf{Y}) \\ &= \mathbf{X}_t \otimes \mathcal{D}(\mathbf{X}_t, \mathbf{Y})\end{aligned} \tag{16}$$

where $\mathcal{D}(\mathbf{X}_t, \mathbf{Y}) \in \mathbb{R}^{H\times W\times 2}$ denotes the deformation field acted on $\mathbf{X}_t$. For convenience, we will simplify it to $\mathcal{D}$ which is generated pixel by pixel by matching $\mathbf{X}_t$ and the neighboring pixels of $\mathbf{Y}$ at the corresponding position. The value of $\mathcal{D}$ at (i, j) can calculated as follows:

$$\begin{aligned}\mathcal{D} &= \mathrm{Proj}(\mathrm{Max}(\mathcal{D}^{\tau})) \\ &= \mathrm{Proj}\left(\mathrm{Max}\left(\frac{Q^{(i,j)}\left[K^{(i-1:i+1),(j-1:j+1)}\right]^T}{\sqrt{C}}\right)\right)\end{aligned} \tag{17}$$

where $\mathcal{D}^{\tau}$ is the similarity matrix to represent the local similarity between $\mathbf{X}_t$ and $\mathbf{Y}$ at (i, j). Since the input image has already been coarsely registered with CRN, the range of searching neighborhood-matching pixels does not need to be too large. We choose a window size of 3, so $\mathcal{D}^{\tau} \in \mathbb{R}^{3\times 3}$. In (17), $Q^{(i,j)}$ is the query feature of $\mathbf{X}_t$ at (i, j) and $K^{(i-1:i+1),(j-1:j+1)}$ is the key feature of $\mathbf{Y}$ in the 3×3 neighboring field around (i, j).

From this, we have derived a more refined deformation field $\mathcal{D}$, which will be then applied to the LR-HSI $\mathbf{X}_t$ to autonomously correct its spatial distortions, thereby minimizing calculation errors in the subsequent fusion process. Similarly, the registered and denoised $\mathbf{Z}_0^{t-1}$ can be inferred following the same procedure:

$$\begin{aligned}\mathbf{Z}_f^{t-1} &= \mathrm{AFR}\left(\mathbf{Z}_0^t, \mathbf{Y}\right) \\ &= \mathbf{Z}_0^t \otimes \mathcal{D}\left(\mathbf{Z}_0^t, \mathbf{Y}\right).\end{aligned} \tag{18}$$

2) S^2CF Model: When improving the quality of LR-HSIs with the aid of HR-MSIs, the focus is often primarily on increasing spatial resolution by incorporating the refined features from HR-MSIs, which frequently overlooks the potential distortions in spectral information that may occur during this process. Given the relatively aligned LR-HSIs and HR-MSIs obtained from the previous operation using AFR, we aim to propose an enhanced S^2CF strategy to guide the model in prioritizing critical features, enhancing its ability to effectively integrate and process relevant information. To effectively address spatial and spectral information independently, we design two distinct branches in S^2CF, where one branch is dedicated to enhancing spatial details while the other tends to preserve the original spectral prior information as much as possible.

Considering that structure tensors (STs) are effective in extracting edge and texture information from images, adapting to various types and sizes of details, and aiding in noise identification and suppression, we design the spatial enhancement branch for spatial details growth in images by constructing STs. To effectively capture the detailed characteristics in HR-MSI $\mathbf{Y}$ which can be transferred to the LR-HSI, we first apply the Sobel operator to extract gradients in the horizontal and vertical directions of $\mathbf{Y}$, which can be expressed as follows:

$$\begin{aligned}\mathbf{G}_h &= \mathbf{S}_h\mathbf{Y} \\ \mathbf{G}_v &= \mathbf{S}_v\mathbf{Y}\end{aligned} \tag{19}$$

where $\mathbf{G}_h$ and $\mathbf{G}_v$ are horizontal and vertical gradient representations, and $\mathbf{S}_h$ and $\mathbf{S}_v$ are corresponding Sobel operators, respectively. Then, the ST can be calculated

$$\mathrm{ST} = \begin{bmatrix}\mathbf{G}_h^2 & \mathbf{G}_{hv} \\ \mathbf{G}_{vh} & \mathbf{G}_v^2\end{bmatrix}. \tag{20}$$

We show the relationship between the determinant of ST and the structural information of the image, it is given as follows:

$$|\mathrm{ST}|\begin{cases} = 0, & \mathbf{G}_h = \mathbf{G}_v = 0 & \text{(21a)} \\ = 0, & \mathbf{G}_h > \mathbf{G}_v = 0 || \mathbf{G}_v > \mathbf{G}_h = 0 & \text{(21b)} \\ > 0, & \mathbf{G}_h > 0 \ \& \ \mathbf{G}_v > 0 & \text{(21c)}\end{cases}$$

where the subequations in (21) correspond to the flat, boundary, and corner regions.

Since we require tensors to represent the structural information of the entire image, however, the regions defined by (21a) and (21b) where $|\mathrm{ST}| = 0$ will result in the loss of structural features, particularly at the marginal sections, which may adversely affect the improvement of image details. Therefore, we propose the advanced version of the structural tensor (AST) to ensure $|\mathrm{ST}| > 0$ at each position without affecting the properties of the determinant in other cases. The AST can be defined

$$\mathrm{AST} = \mathrm{ST} + 1. \tag{22}$$

Then, the spatial weight matrix $\mathbf{W}_{\mathrm{spa}}$ can be defined

$$\begin{aligned}\mathbf{W}_{\mathrm{spa}} &= \begin{vmatrix}\mathbf{G}_x^2 + 1 & \mathbf{G}_{xy} + 1 \\ \mathbf{G}_{xy} + 1 & \mathbf{G}_y^2 + 1\end{vmatrix} \\ &= \left(\mathbf{G}_x^2 + 1\right)\left(\mathbf{G}_y^2 + 1\right) - \left(\mathbf{G}_{xy} + 1\right)\left(\mathbf{G}_{xy} + 1\right).\end{aligned} \tag{23}$$

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Then, the enhanced spatial feature $\mathbf{Y}_{\text{spa}}$ can be obtained

$$\mathbf{Y}_{\text{spa}} = \mathbf{Y} \odot \mathbf{W}_{\text{spa}}. \tag{24}$$

As for the extraction of spectral prior in LR-HSIs at the tth time step, we feed $\mathbf{H}_t$ which is the channel concatenation of $\mathbf{X}_{t+1}$ and $\mathbf{Z}_f^{(t+1)}$ to the spectral extraction branch. We first use a 3×3 convolution to unify the channel dimension and extract the shallow features f_{spe} through a residual block. Therefore, the spectral attention matrix $\mathbf{W}_{\text{spe}}$ is acquired

$$\mathbf{W}_{\text{spe}} = \text{Linear}(\text{Linear}(f_{\text{spe}})). \tag{25}$$

Following, the enhanced feature $\mathbf{H}_{\text{spe}}$ in the spectral dimension can be defined:

$$\mathbf{H}_{\text{spe}} = \mathbf{W}_{\text{spe}} * f_{\text{spe}}. \tag{26}$$

To fully integrate both the spatial and spectral characteristics, we take $\mathbf{Y}_{\text{spa}}$ as the spatial guiding information which will be transferred and blended to the spectral feature $\mathbf{H}_{\text{spe}}$ through a guided filter. Consequently, we are able to derive the mixed features $\mathbf{F}_m$, which will subsequently facilitate the reconstruction of $\mathbf{Z}_0^{(t)}$ and provide a robust foundation for further processing

$$\mathbf{Z}_0^{(t)} = \text{Conv}_{3\times3}(\text{Res Block}(\text{Res Block}(\mathbf{F}_m))). \tag{27}$$

In the reverse process of PSRF-DiffNet, $\mathbf{Z}(T)_0$ undergoes the reconstruction process through S^{2}CF over T steps, ultimately yielding the desired HR-HSI $\mathbf{Z}_0$.

E. Training Objective

The proposed PSRF-DiffNet aims to reconstruct the target Hr-HSI with a sequence of registration and fusion steps in an end-to-end learning manner. Considering the robustness of outliers such as noise and misaligned features in images and the ability to promote sparsity, which will stabilize network training and enable the network to extract crucial features for accurate fusion, we take L1 loss to train the overall network. Therefore, the consistency constraint used in the process of network training to obtain high-quality HR-HSIs is defined as follows:

$$\mathcal{L}_1 = \frac{1}{N}\sum_{n=1}^{N}\left(\left\|\mathbf{Z}^n - \mathbf{Z}_0^n\right\|_1\right) \tag{28}$$

where N is the number of training samples, $\mathbf{Z}_0^n$ is the predicted HR-HSI with the nth training samples {LR-HSI $\mathbf{X}^n$, HR-MSI $\mathbf{Y}^n$}, and $\mathbf{Z}^n$ is the corresponding ground truth.

III. Experiment

In this section, we evaluate the performance of the proposed PSRF-DiffNet. We compare it with multiple competing algorithms on three simulated datasets and one real-world dataset. Here, we present both the quantitative and qualitative results of all methods. Subsequently, we carry out a series of ablation experiments to assess the effectiveness of distinct modules and strategies within PSRF-DiffNet. This systematic analysis aims to comprehensively demonstrate the advantages and contributions of PSRF-DiffNet.

TABLE I
Simulate Distortion Settings on the PaviaC, Houston, and Chikusei Datasets

Dataset	Distortions							
	Elastic				Turbulence			
	α=0	α=200	α=600	α=1000	λ=2	λ=4	λ=6	λ=8
PaviaC	✓	✓	✓	✓				
Houston	✓	✓	✓	✓				
Chikusei					✓	✓	✓	✓

A. Datasets and Implementation Details

1) Datasets Description: To illustrate the effectiveness of the proposed method, we conduct comparative experiments with several competing methods on three public datasets, namely Pavia Center (PaviaC), Houston, and Chikusei. The PaviaC dataset consists of 102 bands with a size of 960×640, which is cropped into patches with 160×160 pixels for training and testing. The Houston dataset with the size $1280 \times 320 \times 144$ is divided into 8 nonoverlapping patches, of which six patches are used for training and the remaining 2 for testing. Each nonoverlapping patch is divided into 41 patches of size $160 \times 160 \times 144$ with a fixed step size. The size of the Chikusei dataset is $2304 \times 2048 \times 110$. The top area with 1792×2048 pixels is selected as the training data, while the remaining area is used as the test data that is split into 64 patches with the size of 128×128. We use Wald's protocol to generate pairs of LR-HSIs and HR-MSIs for training. The spectral bands of HR-MSIs in PaviaC, Houston, and Chikusei are 4, 4, and 3, respectively. These three datasets are degraded spatially with a factor of 4. Furthermore, to validate the performance of the proposed method in real-world scenarios, we conducted experimental verification on the real dataset XijingU. The Xijing University dataset, collected and named by our team at Xijing University in Xi'an, Shaanxi, China, includes an HSI of size $960 \times 960 \times 137$ and a high-resolution multispectral image of size $3840 \times 3840 \times 3$. During the experiments, we first cropped the HSI and generated simulated multispectral images using the Wald protocol. To match the spatial dimension of LR-HSIs with that of HR-HSIs and HR-MSIs, we adopt bicubic interpolation so that the concatenation of noisy HR-HSIs and LR-HSIs can be regarded as inputs for spatial super-resolution in PSRF-DiffNet.

2) Simulated Details: As shown in Table I, for all simulated LR-HSIs, we introduce the elastic and turbulence distortions to simulate the distortions in the process of image acquisition in reality. To thoroughly evaluate the performance of PSRF-DiffNet, we set varying distortion levels for LR-HSIs. Specifically, for elastic distortions, α is set to 200, 600, and 1000. For turbulence distortions, the ratio of aperture diameter to the fried parameter is set to 2, 4, 6, and 8, progressively increasing the distortion levels from mild to severe. Fig. 4 shows the images after different deformations.

3) Implementation Details: We conducted the experiments with the PyTorch framework and trained on four NVIDIA GeForce RTX 3090 GPUs. The experiments were conducted

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TABLE II
QUANTITATIVE RESULTS OF ALL COMPETING METHODS ON THE PAVIAC DATESET WITH DIFFERENT TYPES OF DISTORTIONS

Methods	Well-Registration				α = 200				α = 600				α = 1000			
	PSNR	SAM	ERGAS	CC	PSNR	SAM	ERGAS	CC	PSNR	SAM	ERGAS	CC	PSNR	SAM	ERGAS	CC
CNMF	33.4398	4.5272	3.5338	0.9651	31.2858	5.2977	4.4920	0.9507	27.8231	6.8322	6.7722	0.8816	25.3564	8.1881	8.8818	0.7906
HySure	33.7118	7.4053	3.6164	0.9661	34.8935	6.0275	3.2510	0.9706	33.1883	8.0428	4.0377	0.9650	28.2740	14.0371	7.0003	0.9443
MTF-GLP	29.2902	5.8025	5.3780	0.9215	29.1108	5.6934	5.5684	0.9173	26.2294	7.0439	7.9660	0.8160	23.8276	8.5464	10.1907	0.6750
Integrated	31.3765	4.4836	5.1470	0.9548	25.5545	8.1064	5.9672	0.9421	23.4213	6.5407	10.3416	0.8904	23.1231	7.8829	11.2629	0.7887
U^2-MDN	39.4888	3.7617	2.2671	0.9844	39.3422	3.7966	2.2933	0.9842	39.3220	3.8313	2.3110	0.9842	39.4371	3.8488	2.2676	0.9844
NonRegSRNet	35.3469	3.0618	4.7764	0.9809	35.2422	3.0928	4.7381	0.9813	32.7585	4.1124	5.1830	0.9775	34.4383	4.9373	3.2193	0.9797
MoG-DCN	41.1086	4.4334	1.5766	0.9869	40.8681	3.4956	1.6205	0.9864	40.8537	3.4844	1.6373	0.9868	40.6729	3.5422	2.0774	0.9862
Ours	**42.7744**	**2.7511**	**1.3156**	**0.9905**	**42.6480**	**2.7757**	**1.2935**	**0.9903**	**42.4766**	**2.8126**	**1.2969**	**0.9904**	**42.2527**	**2.8616**	**1.3164**	**0.9899**

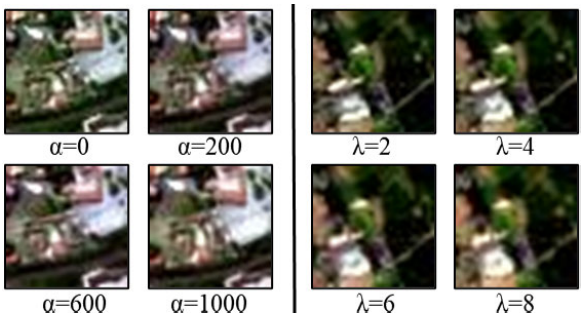


Fig. 4. Visualization results of different methods on the PaviaC datasets with different degrees of elastic deformation.

with a batch size of 8 and 100k iterations on all datasets. The Adam optimizer is employed for the optimization process, with a maximum learning rate set at 0.0001. The time step T is set to 2000, and the hyperparameter sequence $\{\beta_1, \beta_2, \ldots, \beta_n\}$ was defined with uniform growth ranging from 0 to 0.02.

B. Experimental Results

To fully validate the effectiveness of PSRF-DiffNet in this article, we compare it with other methods: CNMF [14], u^2-MDN [52], NonRegSRNet [53], HySure [55], MTF-GLP [10], and MoG-DCN [56]. Among them, CNMF, HySure, MTF-GLP, and Integrated are model-based traditional methods, and the rest are DL-based methods.

We conducted systematic experimental validation on three public datasets and one real-world dataset. For the three public datasets, the fusion performance of different methods was quantitatively analyzed using objective evaluation metrics such as PSNR, CC, SAM, and ERGAS. For the real-world dataset, we employ the quality w/no reference (QNR)

$$\mathrm{QNR} = (1 - D_\lambda)^\alpha (1 - D_S)^\beta \tag{29}$$

where D_λ accesses the spectral distortion and D_S quantifies the spatial distortion. Therefore, QNR provides a comprehensive measurement of quality. While QNR goes higher, D_λ and D_S remain lower, the quality of the fused results gets better. In the results, the highest values are highlighted in bold black, while the second-highest values are indicated with an underline.

1) PaviaC and Houston Datasets: We applied varying levels of elastic distortion to the PaviaC and Houston datasets, with distortion scales α set to 200, 600, and 1000, to evaluate the performance of the proposed method. We first evaluated the visual quality of the fused images generated by PSRF-DiffNet. Figs. 5 and 6 present the fused images and residual images from PaviaC and Houston, respectively, where $\alpha = 0$ and 600 are the original images and $\alpha = 200$ and 1000 are the residual images. The figure shows that well-registered images generally achieve optimal fusion results. As the degree of deformation increases, the adaptability of classical methods declines significantly, resulting in a sharp decrease in fusion quality. Deep learning-based methods are also affected to varying degrees; for example, the method of NonRegSRNet exhibits notable errors at object boundaries. Although MoG-DCN and u2MDN demonstrate higher robustness across different levels of deformation, their fusion quality still does not surpass the method proposed in this article. To validate the fusion performance of the proposed method in regions with fine details, we magnified local areas of the subjective images by a factor of 2 to more clearly observe the improvement of the proposed method in detailed regions. Equally, it can be observed that the proposed method exhibits significant advantages in local regions, effectively preserving spatial details and spectral consistency. This fully highlights the superiority of the proposed method compared to other approaches. Table II summarizes the objective results obtained using the PaviaC dataset under different conditions. In the table, the bolded black font indicates the best results, while the underlined ones represent the second-best results. The results show that deep learning-based methods, as well as the proposed method, achieve better reconstruction accuracy than classical methods under well-registered conditions. However, as deformation increases, the performance of Integrated and NonRegSRNet methods gradually deteriorates. Although MoG-DCN is less sensitive to deformation and demonstrates good stability, a comparison across all methods shows that the proposed method is the most stable, delivering optimal performance under various deformation conditions. Table III shows the objective results for the Houston dataset, where the correlation coefficient (CC) achieves the second-best at $\alpha = 600$, while other results achieve the best outcomes.

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TABLE III
QUANTITATIVE RESULTS OF ALL COMPETING METHODS ON THE HOUSTON DATASET WITH DIFFERENT TYPES OF DISTORTIONS

Methods	**Well-Registration**				α = **200**				α = **600**				α = **1000**			
	PSNR	SAM	ERGAS	CC	PSNR	SAM	ERGAS	CC	PSNR	SAM	ERGAS	CC	PSNR	SAM	ERGAS	CC
CNMF	38.5277	5.0334	3.1643	0.9605	38.4165	5.1236	3.2212	0.9606	35.6728	6.5387	3.2212	0.9192	33.2087	7.9574	5.8916	0.8608
HySure	37.2311	5.6264	4.3269	0.9296	36.1447	6.2411	4.7345	0.9395	36.6587	5.8414	4.3950	0.9320	34.3724	7.5768	5.5402	0.9203
MTF-GLP	37.4648	4.0079	3.4866	0.9520	37.1801	4.1610	3.5990	0.9495	34.2103	5.6892	5.0801	0.8935	32.1783	7.0761	6.3943	0.8973
Integrated	38.3956	3.6480	3.2069	0.9627	37.1296	4.2946	3.9197	0.9450	34.9198	5.4761	4.9841	0.9039	33.5003	6.4843	5.8577	0.8554
U^2-MDN	44.3270	2.5428	1.8717	0.9855	44.0988	2.6458	1.9331	0.9850	43.9087	2.6952	1.9690	0.9849	44.1351	2.6263	1.9317	0.9851
NonRegSRNet	40.2726	2.9540	2.7185	0.9826	40.0864	3.1099	2.7902	0.9834	39.7181	3.1158	2.8874	0.9833	39.8896	2.9160	2.8003	0.9847
MoG-DCN	50.7238	1.3160	1.1779	0.9914	48.9844	1.6449	1.3558	0.9888	47.4965	1.9180	1.5500	**0.9880**	46.1357	2.4216	1.7684	0.9832
Ours	**51.0654**	**0.9910**	**0.8922**	**0.9918**	**49.3239**	**1.1251**	**1.0139**	**0.9907**	**48.0778**	**1.7092**	**1.1908**	0.9879	**47.4320**	**1.5348**	**1.2240**	**0.9857**

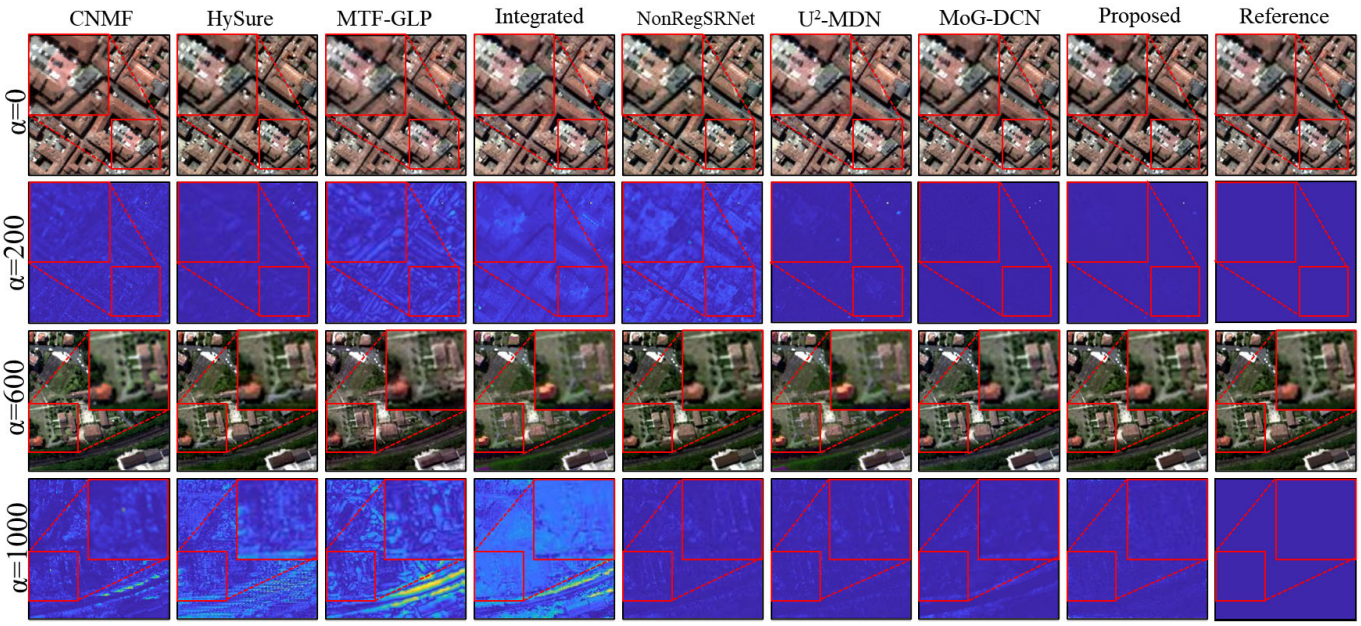


Fig. 5. Visualization results of different methods on the PaviaC dataset with different degrees of elastic deformation.

2) Chikusei Dataset: To validate the robustness of the proposed method against other types of deformations, we added turbulence distortions to the Chikusei dataset to further assess its performance. The deformation scales λ are set to 2, 4, 6, and 8, respectively. In contrast, the proposed method not only effectively preserves edge and detail information under various deformation conditions but also maintains overall visual consistency, demonstrating high visual robustness, the subjective results are shown in Fig. 7. From the zoomed-in visualization results, the proposed method maintains excellent spatial fidelity even under turbulent distortion conditions, significantly outperforming comparative methods. Additionally, as seen in Table IV, traditional methods continue to perform stably under turbulence deformation. Integrated is more sensitive to turbulence noise, with fusion accuracy decreasing significantly as deformation intensity increases, while NonRegSRNet is less affected. Although MoG-DCN and u2MDN show greater robustness, their accuracy still falls short of the proposed method.

3) XijingU Dataset: To thoroughly validate the performance of the proposed PSRF-DiffNet in real-world scenarios, we conducted additional experiments on the real dataset XijingU. Considering the complexity and diversity of deformations in real datasets, we added varying degrees of elastic deformation and turbulence deformation to the HSIs for mixed training, followed by testing with real hyperspectral and multispectral image pairs. As shown in Fig. 8, we present a comparison of the visualization results of different methods on the XiJingU dataset. It can be observed that there are significant differences in the image-processing performance among the methods. Specifically, the comparative methods exhibit certain limitations when handling complex scenes, particularly in terms of detail preservation and edge clarity, making it difficult to fully restore high-quality image information. In contrast, our proposed PSRF-DiffNet demonstrates exceptional performance on real datasets. As shown in Table V, PSRF-DiffNet significantly outperforms other comparative methods on the QNR metric, achieving the best performance. This result fully

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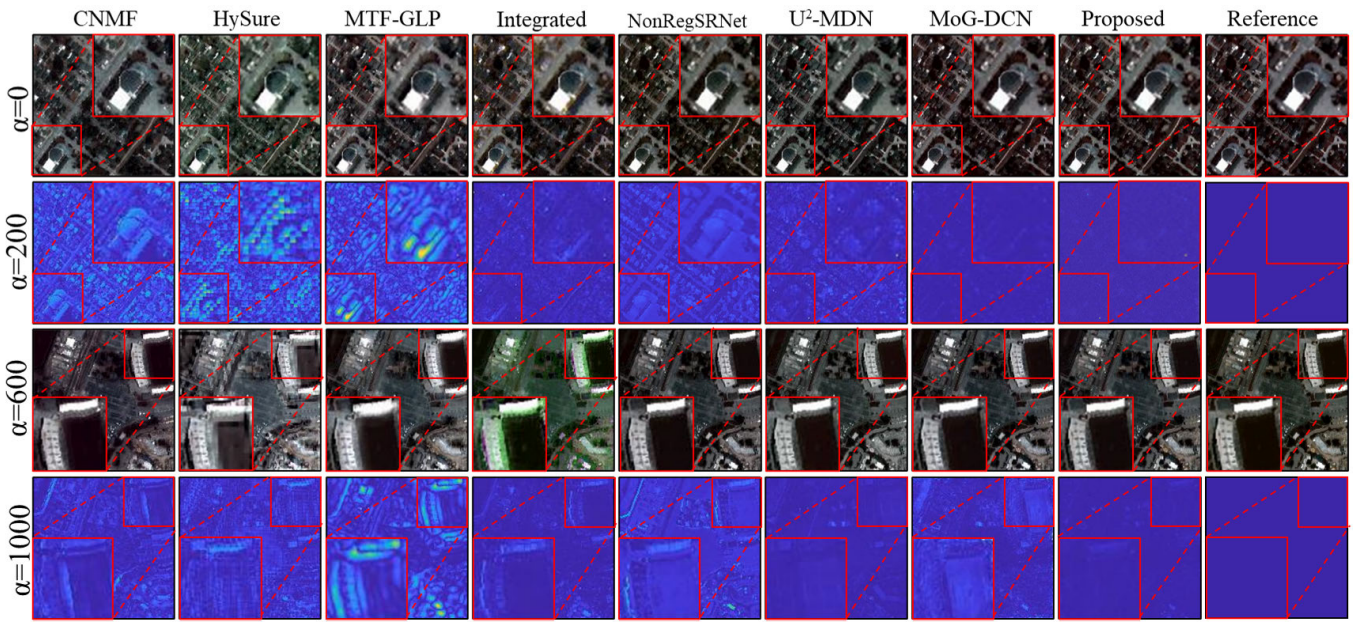


Fig. 6. Visualization results of different methods on the Houston dataset with different degrees of elastic deformation.

TABLE IV
QUANTITATIVE RESULTS OF ALL COMPETING METHODS ON THE CHIKUSEI DATASET WITH DIFFERENT TYPES OF DISTORTIONS

Methods	$\lambda = 2$				$\lambda = 4$				$\lambda = 6$				$\lambda = 8$			
	PSNR	SAM	ERGAS	CC	PSNR	SAM	ERGAS	CC	PSNR	SAM	ERGAS	CC	PSNR	SAM	ERGAS	CC
CNMF	31.9740	3.4353	4.9714	0.8969	31.5939	3.4800	5.2033	0.8899	31.3192	3.6755	5.6392	0.8798	30.9099	3.8698	5.9617	0.8698
HySure	25.2948	5.4365	6.5747	0.7367	25.4888	5.3797	6.2986	0.7504	25.7959	5.2249	6.0079	0.7644	26.1673	5.0336	5.7513	0.7776
MTF-GLP	32.1415	3.2745	5.5447	0.8954	31.9305	3.2985	5.6876	0.8887	31.6469	3.3936	5.5916	0.8799	31.2959	3.5471	6.2417	0.8681
Integrated	26.3952	2.4519	5.9446	0.9461	25.5089	2.8838	6.8237	0.9125	24.6655	3.2195	7.4608	0.8895	24.0278	3.4993	8.0439	0.8693
U²-MDN	33.3864	3.5333	4.7070	0.9175	33.0169	4.1021	5.0558	0.9105	32.7793	4.1918	4.9973	0.9088	31.1025	4.1443	5.2067	0.9096
NonRegSRNet	36.5768	2.3293	2.3468	0.9852	36.1984	2.2959	3.2095	0.9575	35.5783	1.9179	3.2128	0.9595	34.3125	2.0954	3.80776	0.9565
MoG-DCN	40.3438	1.1264	2.0238	0.9880	37.0135	1.8488	4.6868	0.9874	37.7668	2.2876	1.3718	0.9864	31.9579	2.9524	8.4236	**0.9859**
Ours	**41.0324**	**1.3856**	**0.9796**	**0.9909**	**40.7354**	**1.4086**	**1.0557**	**0.9898**	**39.6110**	**1.4888**	**1.0831**	**0.9889**	**38.9195**	**1.5650**	**1.2215**	0.9854

validates the robustness and effectiveness of PSRF-DiffNet in handling complex scenes, highlighting its practical value in real-world scenarios.

4) Spectral Fidelity Performance: We also demonstrate the spectral curves of various land covers on the Houston and Chikusei datasets. The Houston dataset includes a range of land covers such as road, roof, lawn, and tree, while the Chikusei dataset includes house, grass, path, and soil. Fig. 9 illustrates the spectral curve results of our proposed method compared to the reference method. It is evident that PSRF DiffNet is closer to the reference spectral curves of eight different land cover types, while other comparison methods have varying degrees of deviation from the reference spectral curves. This result confirms that our method can effectively preserve the original spectral information of low-resolution HSIs, enabling the reconstruction of high-resolution HSIs with high spectral fidelity.

5) Registration Performance: To quantitatively evaluate the registration performance of this method, we have conducted additional experiments to evaluate the registration accuracy of all data fusion methods. The PSRF-DiffNet employs an end-to-end joint optimization framework to achieve simultaneous modeling of registration and fusion. Since there are no explicit intermediate registration results, the registration accuracy is indirectly evaluated by quantifying the deformation magnitude between the fused results and the reference image. Specifically, we have introduced two widely used metrics, MI and root mean square error (RMSE), to quantitatively assess the registration quality. The experimental results (Table VI) demonstrate that our proposed method achieves superior registration accuracy compared to other fusion methods, as evidenced by higher MI values and lower RMSE values. These results fully demonstrate the strong adaptability of the proposed PSRF-DiffNet for registration and fusion tasks.

C. Ablation Studies

In this section, we perform ablation studies on the PaviaC ($\alpha = 600$), Houston ($\alpha = 1000$), and Chikusei ($\lambda = 4$) datasets.

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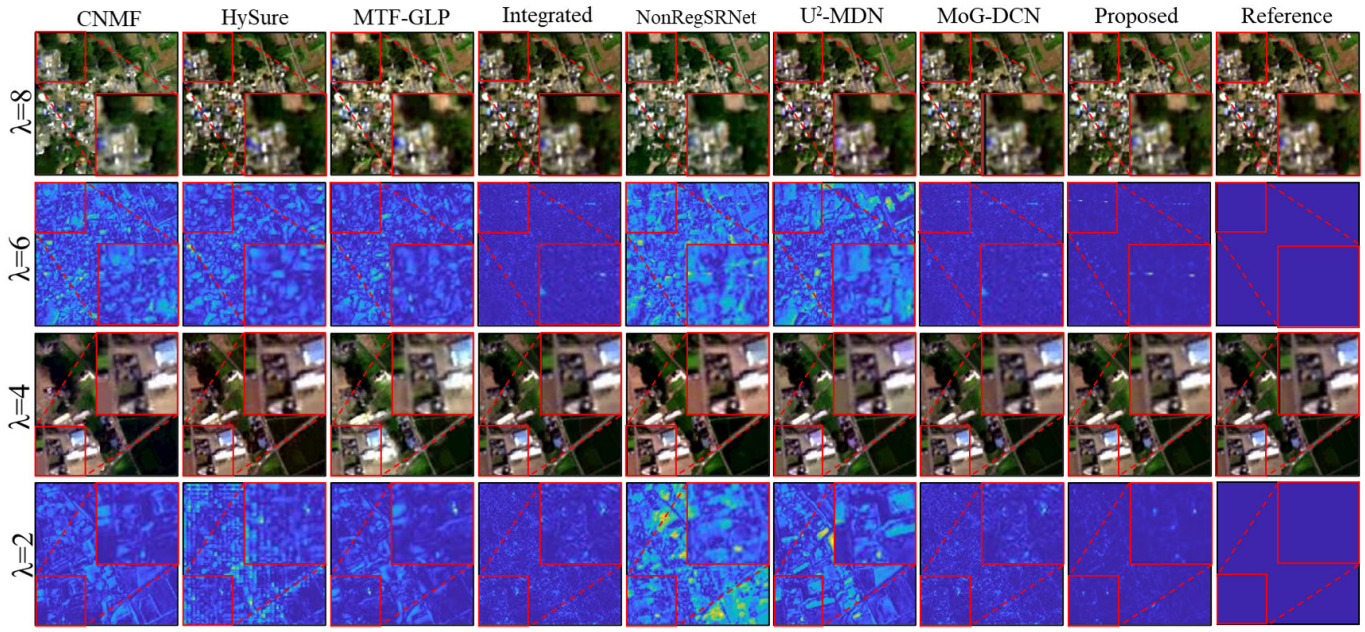


Fig. 7. Visualization results of different methods on the Chikusei dataset with different degrees of elastic deformation.

TABLE V
QUANTITATIVE COMPARISON ON THE XIJINGU DATASET

Method	CNMF	HySure	MTF-GLP	Integrated	u^2-MDN	NonRegSRNet	MoG-DCN	Ours
QNR	0.6114	0.7152	0.6082	0.7231	0.6905	0.5492	0.7715	**0.8160**
D_λ	0.3064	0.2065	0.2965	0.2315	**0.1271**	0.1391	0.2085	0.1350
D_S	0.1318	0.1078	0.1317	0.0932	0.2103	0.3693	**0.0255**	0.0552

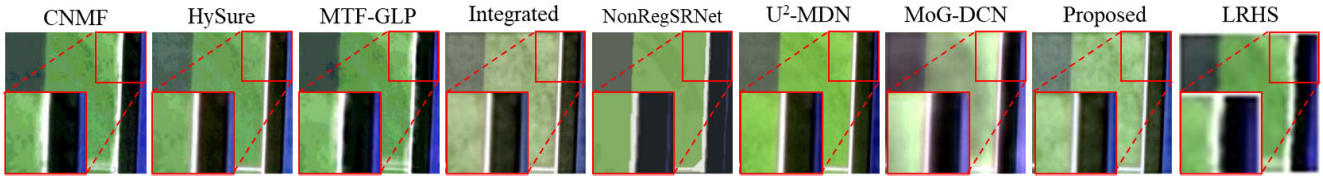


Fig. 8. Visualization results of different methods on the XiJingU dataset.

TABLE VI
EVALUATION OF THE REGISTRATION PERFORMANCE OF OUR METHOD AND COMPARISON METHODS ON DIFFERENT DATASETS

Methods	Pavia		Houston		Chikusei	
	α=600		α=1000		λ=8	
	MI	RMSE	MI	RMSE	MI	RMSE
CNMF	2.4528	0.0509	2.1590	0.0373	2.2886	0.0413
HySure	2.9910	0.0443	1.8327	0.0235	2.3135	0.0536
MTF-GLP	1.4969	0.0382	1.9538	0.0182	2.1696	0.0339
Integrated	1.7352	0.0736	1.7298	0.0219	1.4966	0.0664
u^2-MDN	4.7286	0.0103	3.4135	0.0063	2.5526	0.0301
NonRegSRNet	3.8822	0.0259	2.3899	0.0093	1.4976	0.0773
MoG-DCN	4.6655	0.0097	3.4026	0.0049	**4.0130**	0.0148
Ours	**5.0542**	**0.0081**	**4.0704**	**0.0043**	3.9352	**0.0117**

1) Strategy of Progressive Synergistic Registration and Fusion: In the proposed PSRF-DiffNet, we perform multilevel iterative fusion and registration during the reverse process of the diffusion model to generate high-quality fused images. To thoroughly validate the effectiveness of this iterative strategy, we implemented a single-iteration variant, PSRF-SingleNet, for performance evaluation. In PSRF-SingleNet, we obtained the fusion result through a single registration and fusion step. The results shown in Fig. 10 indicate that PSRF-DiffNet consistently outperforms PSRF-SingleNet across all datasets.

2) FRFB Model: FRFB integrates registration and fusion at each time step, aiming to gradually reduce the errors in registration and fusion throughout the entire iterative process. To thoroughly validate the effectiveness of the proposed structure, we designed a variant structure, PSRF DiffNet-Unet, by replacing the FRFB in PSRF DiffNet with the classic Unet structure. As shown in Fig. 11, our proposed PSRF DiffNet demonstrated superior performance, with all indicators surpassing the variant PSRF DiffNet-Unet, indicating that our FRFB can effectively enhance the quality of image reconstruction.

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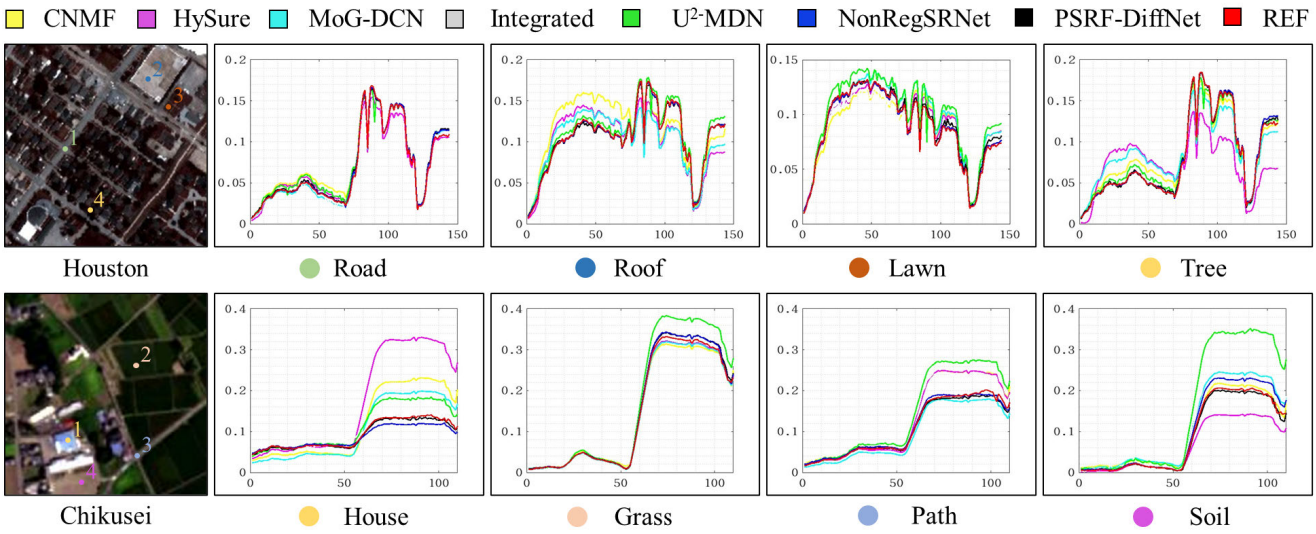


Fig. 9. Comparison of spectral fidelity performance of the competing methods on different ground objects in the Houston and Chikusei datasets.

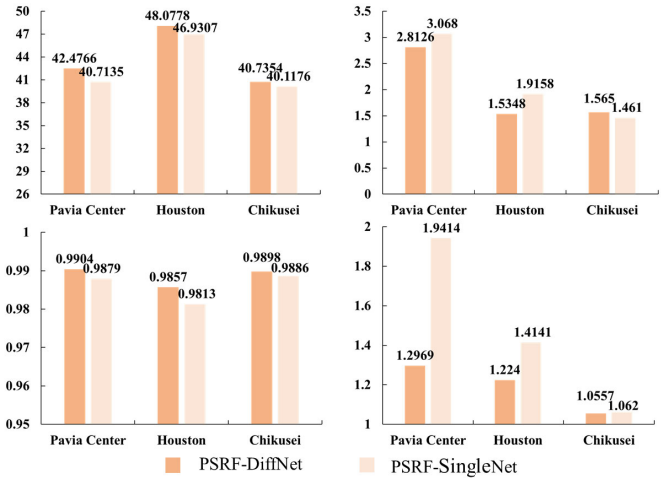


Fig. 10. Comparison of experimental results between the PSRF-DiffNet and PSRF-SingleNet.

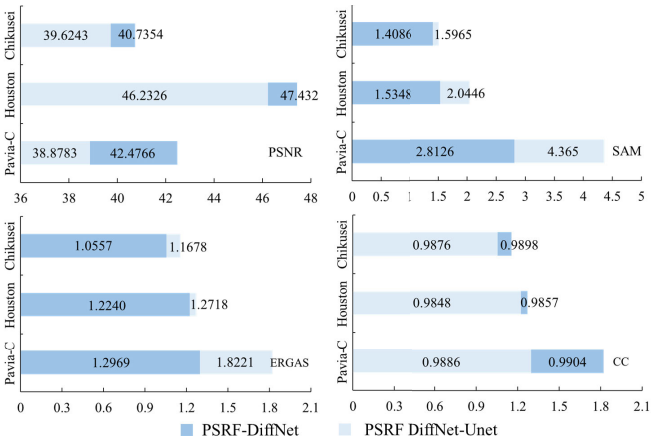


Fig. 11. Comparison of experimental results between the PSRF-DiffNet and PSRF-UNet.

3) Coarse Registration Network: To reduce the complexity of the registration process in inverse operations, we have introduced a CRN, which is responsible for the preliminary alignment of the initial image pair. To validate the effectiveness of CRN, we constructed a comparative model named PSRF-DiffNet-NoCRN, which does not include the CRN. Table VII summarizes the comparative experimental results between our proposed network and its variants. The data in Table VII clearly demonstrate that our proposed method outperforms other alternatives, highlighting the CRN's capability to perform effective initial alignment and produce high-quality fusion results.

TABLE VII
COMPARISON OF EXPERIMENTAL RESULTS BETWEEN PSRF-DIFFNET AND PSRF-UNET-NOCRN

Variant	Dataset	PSNR	SAM	ERGAS	CC
PSRF-DiffNet	PaviaC	42.4766	2.8126	1.2969	0.9904
	Houston	47.4320	1.5348	1.2240	0.9857
	Chikusei	40.7354	1.4086	1.0557	0.9898
PSRF-DiffNet-NoCRN	PaviaC	40.0223	3.3120	1.7229	0.9855
	Houston	46.3875	1.9039	1.4969	0.9826
	Chikusei	39.5473	1.4866	1.1598	0.9873

IV. CONCLUSION

In this article, through deeply investigating the interdependence between image fusion and registration, we propose a PSRF-DiffNet. This network adopts a registration strategy from coarse to fine and utilizes a diffusion model for iterative optimization to achieve precise image registration and fusion. The proposed method enables joint optimization of registration and fusion with tolerance for partial misalignment, effectively reducing the impact of registration errors on fusion results, and it adapts to diverse deformations through a combination of coarse-to-fine rigid and nonrigid registration, ensuring robustness in complex environments. Extensive experiments on three datasets demonstrate the effectiveness of the proposed method and the ability of the network to produce high accuracy and stable reconstructed images under complex nonrigid deformation conditions. Since we perform an iterative optimization

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process, the computational cost might be relatively high, especially when dealing with large-scale datasets. In light of the identified areas for improvement and the evolving demands within the field of unregistered image fusion, our subsequent research endeavors will be centered on the proposition of a novel lightweight paradigm to significantly reduce the computational load and ensure optimal performance.

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